AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 2 · STUDENT LESSON

Input, Process, Output

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 1: a visible output does not reveal the decision mechanism.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace information through a system without confusing information with a physical action. Try the recall before opening notes.

Your learning target

I can **trace** information through a system without confusing information with a physical action.

I can explain which details a **representation** preserves or discards.

I can locate the earliest stage at which an incorrect result could have arisen.

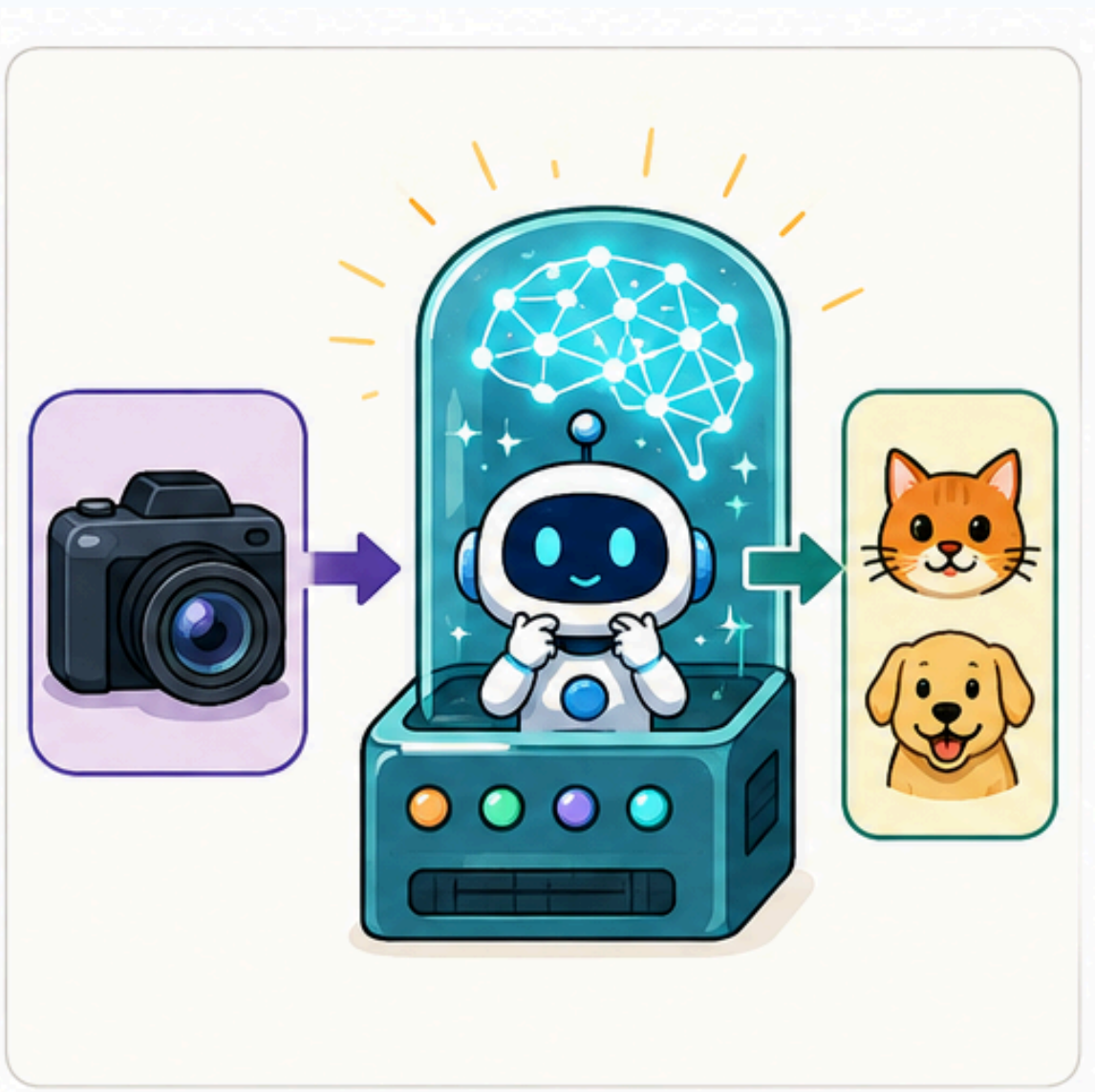
Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A system transforms information; it does not simply 'know.'



Your target

LEARNING TARGET A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud.

> Which stage contradicts that claim?

Learner name:

Readiness for the connected explanation

> In last week's bus case, the forecast number and the display colour came from different decisions. Which decision produced each **output**?

This week’s end goal

By the end, I can **trace** an information path and name what a summary loses.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Input, Process, Output case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A music helper receives listening choices, compares them with a ranking **process**, and suggests a song. The suggestion appears only after information has moved through the system.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A library kiosk asks Sam for a genre. Sam selects mystery. The kiosk compares that preference with catalogue tags and displays three book titles. Sam says the kiosk has read three books aloud.

That conclusion adds an action that the description never supplied: displaying a title is different from producing spoken audio.

An information path names what enters each stage, what happens there and what leaves it. Treat **input** and **output** as roles relative to a stage.

A list of titles is the **output** of the search, but it could become the **input** to a separate text-to-speech stage. A component can therefore receive the **output** of another component without doing the same job.

The kiosk's genre-matching operation can be implemented with a **written rule**. Nothing in this description establishes that a model was trained.

This connects to Week 1: tracing information tells us what a **process** needs to do, while **mechanism evidence** tells us how it does it.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Input, Process, Output. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



INPUT

songs played, liked, or skipped



PROCESS

compare clues and rank possible songs



INPUT songs played, liked, or skipped

PROCESS compare clues and rank possible songs

OUTPUT a suggested song or a not-sure result

Key idea

CORE IDEA A system transforms information; it does not simply 'know.'

How the mechanism works

UNDER THE HOOD **Input**, **process**, and **output** are functional roles. The **process** may contain human choices, fixed rules, calculations, retrieval, or a **learned model**.

Naming the three stages prevents magical explanations of computer behavior.

Draw a fresh example with all three stages:

A closer explanation

Two possible library information paths

INPUT	PROCESS	OUTPUT
Genre request mystery and catalogue tags	Keep matching catalogue records	Three matching titles
Three titles from the first stage	A separate speech system if available	Spoken titles rather than whole books
A selected title and an available copy	A person completes the borrowing process	A recorded loan and a borrowed copy

The first row is the kiosk as described. The later rows require additional components and actions. Arrows show information dependencies; they do not establish that every stage has actually run.

Notice the difference between a title, a description and an entire book. All three can be text, but they represent different amounts of information.

Even a speech stage receiving three titles can speak only those supplied titles; it has not thereby read the full books. Always identify the content of an **input**, not just its data type.

A recommendation is also different from a loan. Selecting a title does not reserve a copy unless an authorised reservation step exists.

Keeping these distinctions clear prevents us from crediting a system with actions it did not take.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

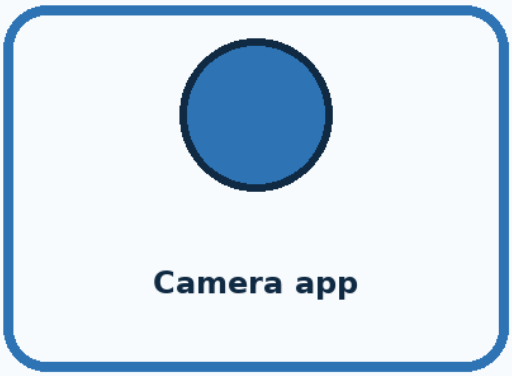
Compare the cases

> Compare the supplied cases for Input, Process, Output. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Camera app Clue: photo enters before any result. Meaning: **input**.

The case to investigate

CASE 2 — Music ranking Clue: choices are compared and scored. Meaning: **process**.

The case to investigate

CASE 3 — Song suggestion Clue: a result is shown to the user. Meaning: **output**.

The case to investigate

CASE 4 — No listening history Clue: important **input** is missing. Meaning: not-sure case.

Keep this boundary clear

BOUNDARY The **output** is not proof that the system understood the **input** as a person would.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Store red count alongside total.

Predicted result

Prediction: The records become (6,2) and (6,5).

Why the result follows

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

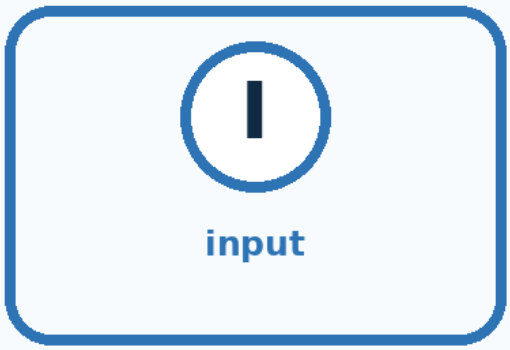
Words that unlock the idea

> Use the vocabulary of Input, Process, Output to explain a concrete decision, rather than only memorising definitions.

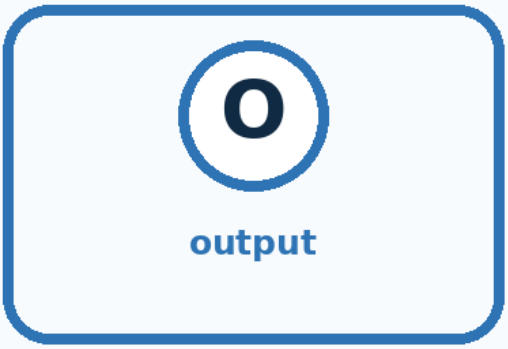
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
input	information entering a system	
process	the steps that transform information	
output	the result produced by the system	
trace	follow information through each stage	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR The computer already knows the answer before it receives **input**.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Representation	A way of recording information so that it can be stored, communicated or processed
Information loss	Details that cannot be recovered from the retained representation

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STUDENT LESSON · 6 OF 18

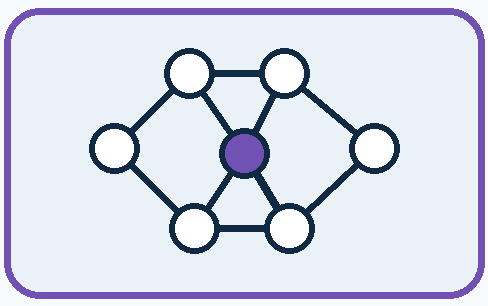
Follow a worked example

> Follow the worked trace for Input, Process, Output, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A music helper receives listening choices, compares them with a ranking **process**, and suggests a song. The suggestion appears only after information has moved through the system.

Step 1

STEP 1 — OBSERVE songs played, liked, or skipped

Step 2

STEP 2 — **TRACE** compare clues and rank possible songs

Step 3

STEP 3 — CONCLUDE a suggested song or a not-sure result

Model answer

MODEL REASONING Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the **process** is unknown, state what **evidence** is missing instead of inventing a **mechanism**.

A claim to avoid

DO NOT OVERCLAIM The **output** is not proof that the system understood the **input** as a person would.

What new **evidence** could change this answer?

A closer explanation

Worked explanation · Input, Process, Output

Consider two catalogue records. Book A has 90 pages and is about a child solving a seaside mystery. Book B has 90 pages and is about a lost spaceship. If we keep only page count, both become the number 90.

The number is useful for a length limit, but it cannot tell us which story is a mystery. Information that was never retained cannot be recovered just by calculating harder.

Representation kept	Question it can help answer	Question it cannot settle alone
Page count 90	Is the book under 100 pages?	Is its story a mystery?
Genre tag mystery	Does it match a mystery request?	Will this particular reader enjoy it?
Short summary	What does the description say happens?	What exact sentence occurs on page 42?

Worked explanation · Input, Process, Output

The phrase cannot settle alone matters. A genre tag might be inaccurate, and a reader might enjoy an unfamiliar genre. A **representation** is useful only in relation to a question and its limits.

More detail is not automatically better: unnecessary personal information can add risk without helping the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Input, Process, Output

A fictional classroom cupboard records coloured pencils, then reports just a total.

The facts we will use

> **Supplied facts**
>
> Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

> **Supplied facts**
>
> The **process** adds colour counts and stores only the total, not the two counts.

AI ACADEMY / AI-1/2

Different colour counts, same total

	Red	Blue	Total
Box A	2	4	6
Box B	5	1	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/2

Input, Process, Output

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read colours

A: red 2, blue 4. B: red 5, blue 1.



2. Add counts

A: 2+4=6. B: 5+1=6.



3. Store total

Both records become total 6.



4. Ask a new question

The saved totals cannot tell which box has more red pencils.

CHANGE ONE THING

Store red count alongside total.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read colours	A: red 2, blue 4. B: red 5, blue 1.
2. Add counts	A: 2+4=6. B: 5+1=6.
3. Store total	Both records become total 6.
4. Ask a new question	The saved totals cannot tell which box has more red pencils.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

What this example does not establish

Keep the stated limit

The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Keep the stated limit

Read the labelled picture: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Keep the stated limit

Change one thing: Store red count alongside total. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Input, Process, Output claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Camera app	photo enters before any result	input
Music ranking	choices are compared and scored	process
Song suggestion	a result is shown to the user	output
No listening history	important input is missing	not-sure case

Repair two tempting claims

Challenge claim A

CLAIM A The computer already knows the answer before it receives **input**.

Repair:

Challenge claim B

CLAIM B The screen is the **process** because that is where the result appears.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

The kiosk is asked for mystery books under 100 pages but displays a 140-page mystery. Before changing everything, inspect the intermediate records. Was the maximum entered as 100? Did the catalogue say 140?

Did filtering preserve the wrong record? Did the display show a title different from the filtered result? The first mismatch between the intended and actual path helps locate the problem.

A correct-looking final screen can hide an earlier error. A calculator can correctly multiply a number that was transcribed incorrectly, and a warning rule can correctly use an incorrect forecast.

Write down intermediate values so you can test each connection.

Explain why a system that displays a book title has not necessarily spoken, reserved or read the book

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STUDENT LESSON · 8 OF 18

Practise together

Practise Input, Process, Output with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A system transforms information; it does not simply 'know.'

Use a fresh example to explain **input**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The screen is the **process** because that is where the result appears.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A greenhouse records morning temperatures of 18, 22 and 26 degrees C, then retains only the largest value, 26. Can the stored value prove that the temperature rose steadily? No.

The sequence 22, 18, 26 has the same largest value but includes a decrease. These two sequences form a counterexample to the claim that the largest value determines the order.

To solve this, first state exactly what was kept: one maximum. Then ask what the claim requires: the order of several readings.

Finally construct two possible originals that share the retained value but give different answers to the claim. If both originals fit the record, the record alone cannot decide between them.

> **A record keeps only a visitor count of 3. Give two different arrival schedules consistent with it and name one fact the count cannot reveal**

Practise with less help: Finish the visitor example

Use workbook W2 for the connected example "**Representation**s and information paths". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Representation**s and information paths. Proposed explanation: "Every **output** is the final action". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

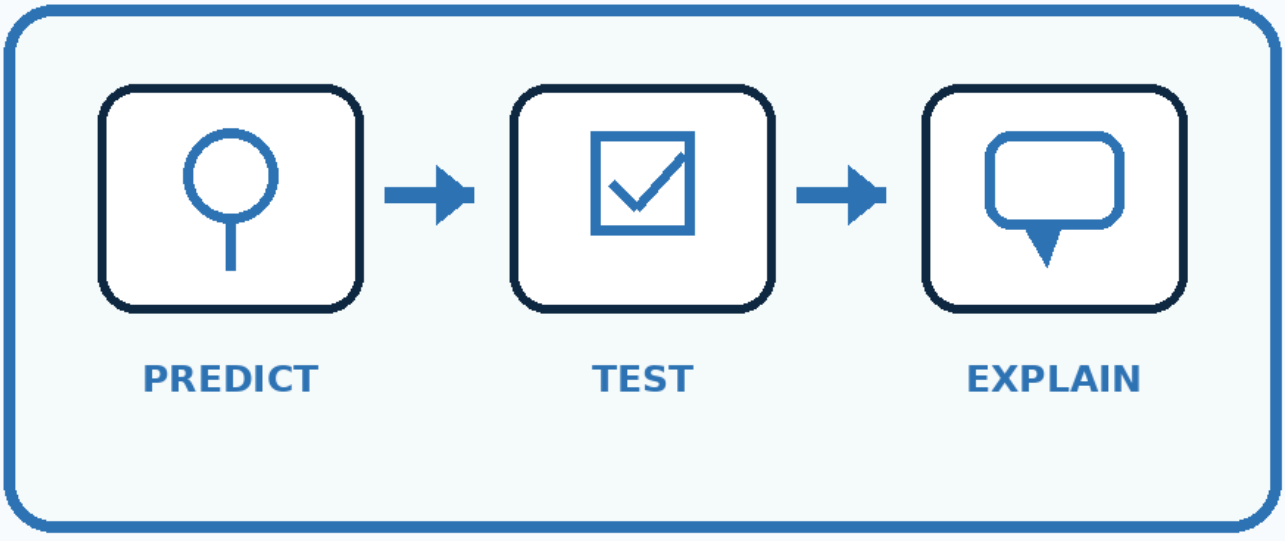
Run the investigation

> Investigate Input, Process, Output. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Human Music Recommender

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: **Trace** a music recommender and draw all three stages.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

STUDENT LESSON · 10 OF 18

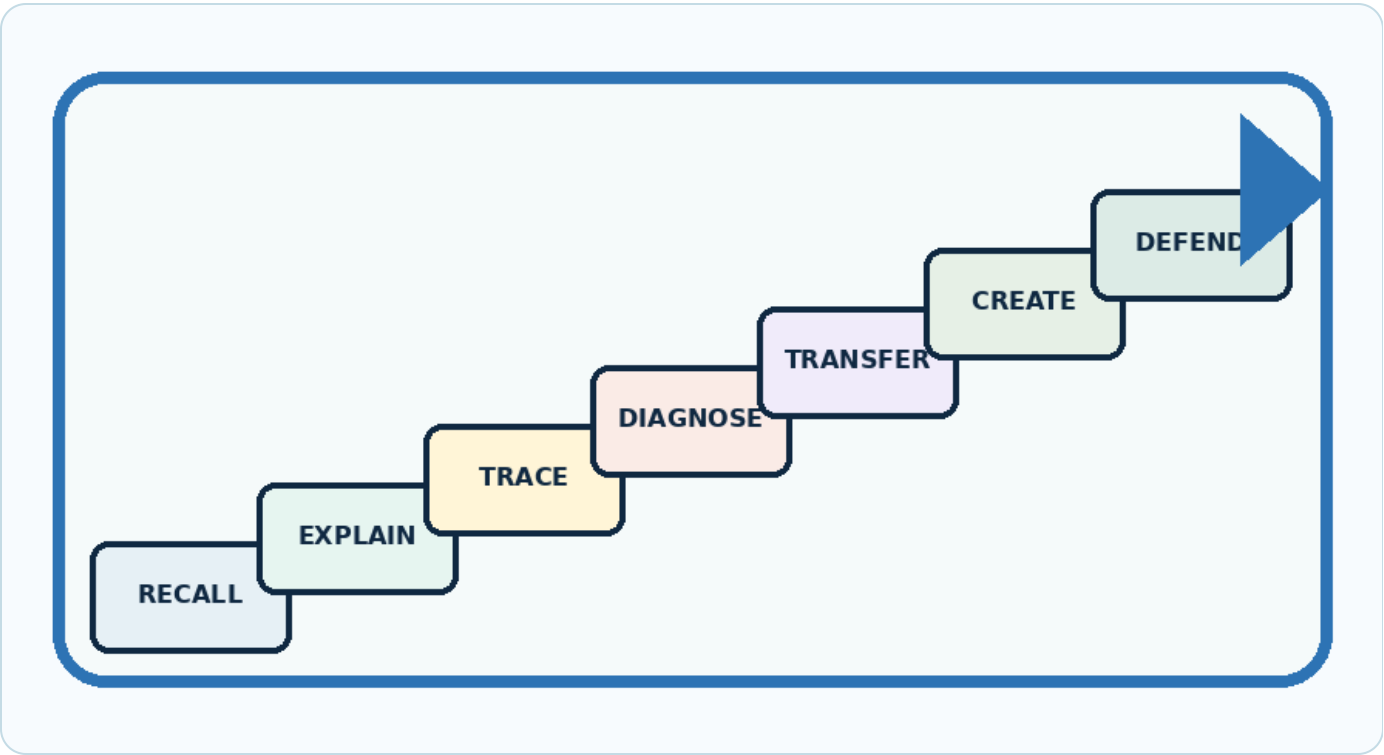
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Input, Process, Output. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **input**.

EXPLAIN A system transforms information; it does not simply 'know.'

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: The computer already knows the answer before it receives **input**.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

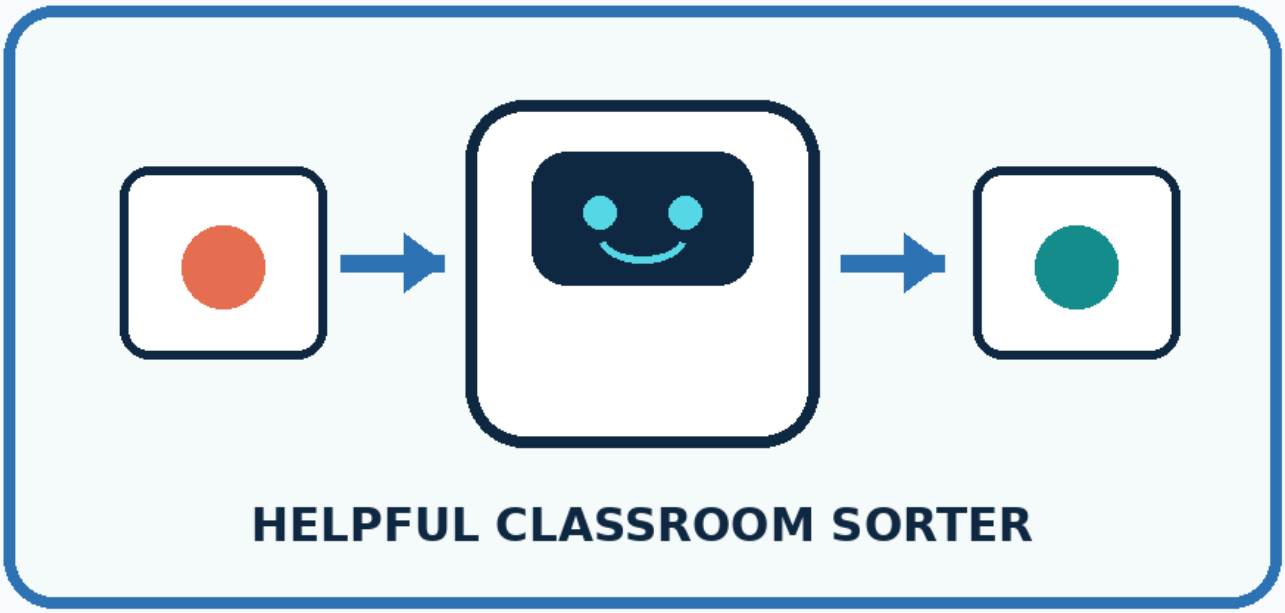
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Draw the Classroom Sorter's first **input-process-output** map and show where a person may review the result.

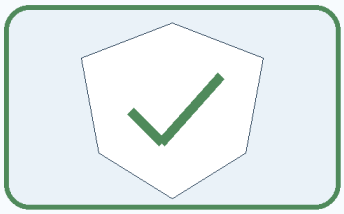
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN The **output** is not proof that the system understood the **input** as a person would.

My project decision:

Evidence for your project

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

STUDENT LESSON · 12 OF 18

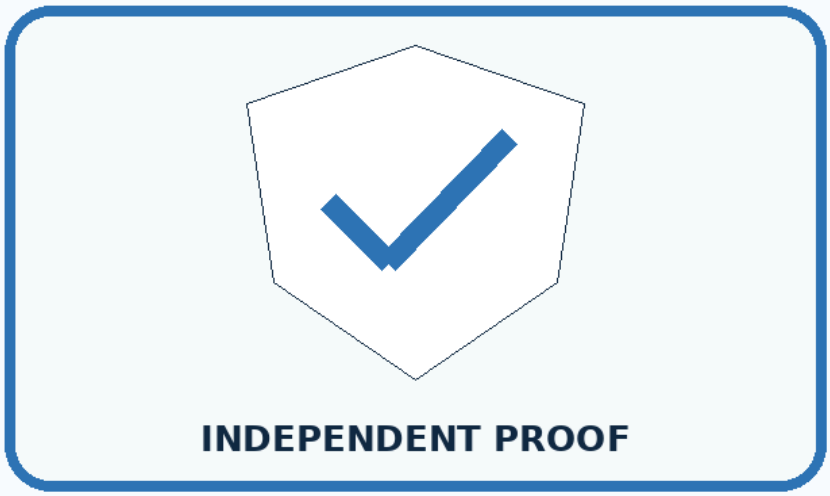
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF **Trace** a fresh system with an accurate **input**, **process**, and **output**, then

explain one missing-**input** case.

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the **process** is unknown, state what **evidence** is missing instead of inventing a **mechanism**.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Sensors Are Messengers.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Input, Process, Output case for supported practice. Repeated examples do not become fresh assessments.

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A playlist helper receives a mood, matches song tags and displays three titles.

The question

> Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Worked explanation

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

From mood to song titles

Evidence or stage	Value or result
Mood	Input from listener

Evidence or stage	Value or result
Tag matching	Process chooses matches
Three titles	Displayed output
Audio playback	A separate possible step

A playlist helper receives a mood, matches song tags and displays three titles.

> Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Investigate a new loss of information

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

My answer, with a **trace**, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Draw the Classroom Sorter's first **input-process-output** map and show where a person may review the result.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Input, Process, Output, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional classroom cupboard records coloured pencils, then reports just a total.

By the end, I can	How I will show it
Trace information through a system without confusing information with a physical action	Explain the deciding step, show a trace or test, and state one limit.
Explain which details a representation preserves or discards	Explain the deciding step, show a trace or test, and state one limit.
Locate the earliest stage at which an incorrect result could have arisen	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Input, Process, Output without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: **Trace** information through a system without confusing information with a physical action

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom cupboard records coloured pencils, then reports just a total.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The saved totals cannot tell which box has more red pencils.

1. Add counts

2. Read colours

3. Ask a new question

4. Store total

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ask a new question The saved totals cannot tell which box has more red pencils.

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Store red count alongside total. Which conclusion is supported by the supplied facts?

1. The records become (6,2) and (6,5).

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Repair a missing building block

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

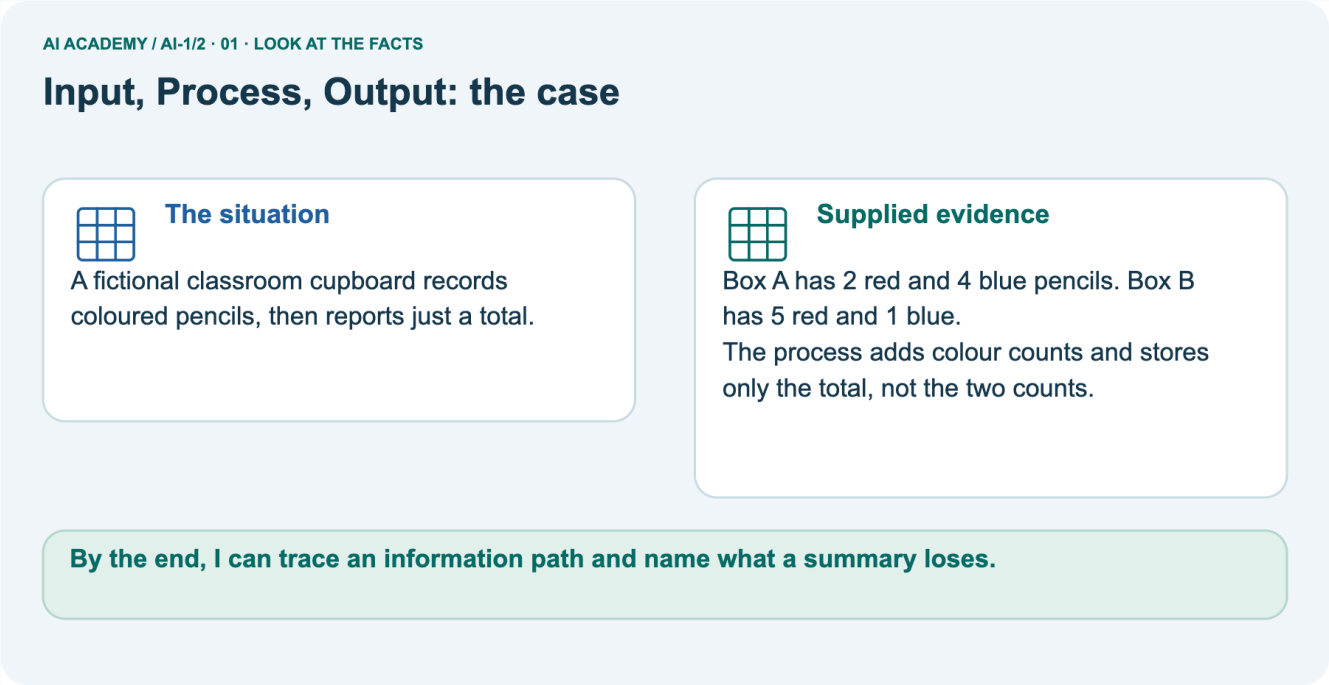
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Input, Process, Output graphic using its labels, deciding evidence and relevant limitation.

01. Input, Process, Output: the case



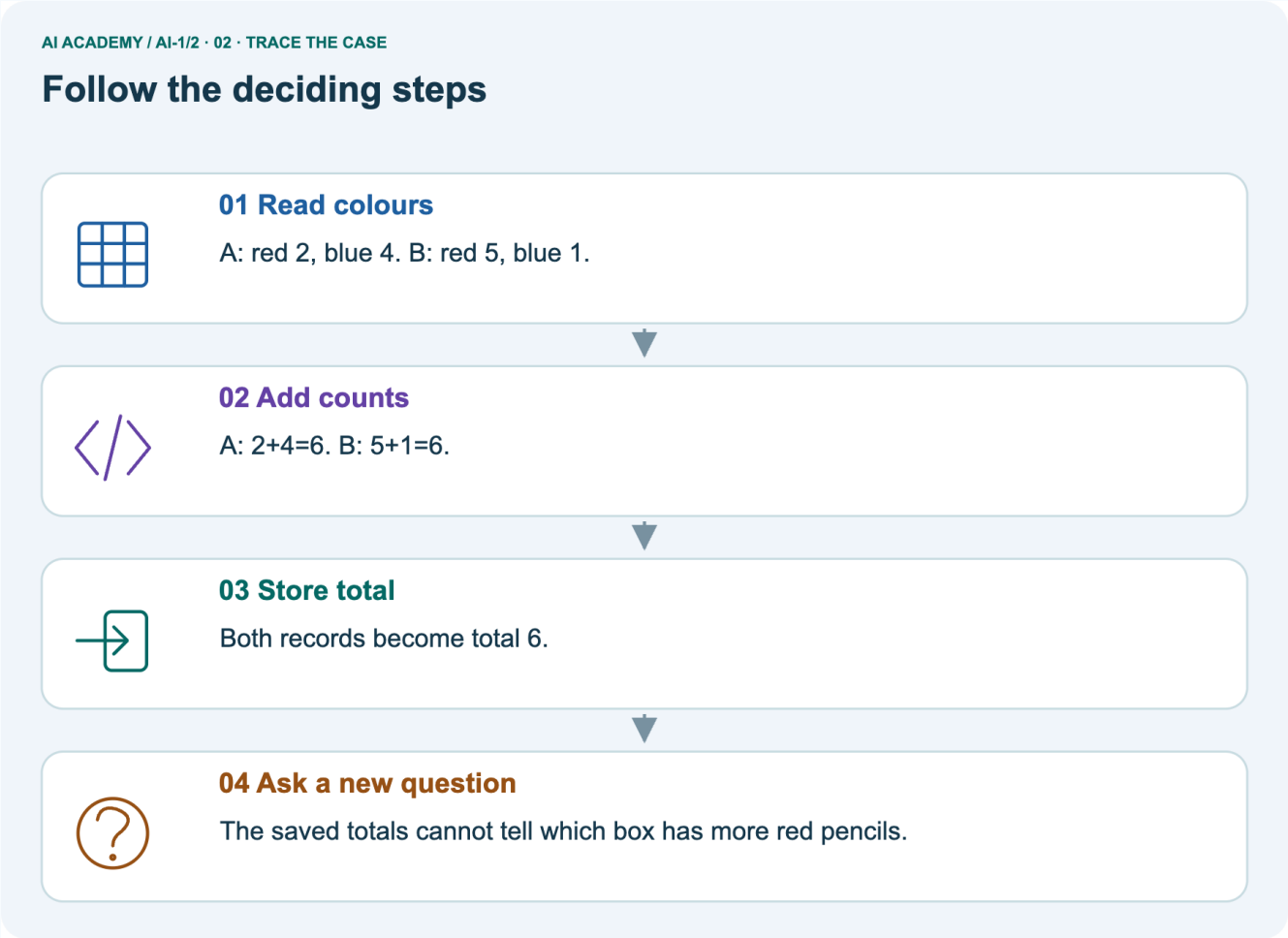
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom cupboard records coloured pencils, then reports just a total. Supplied facts: Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

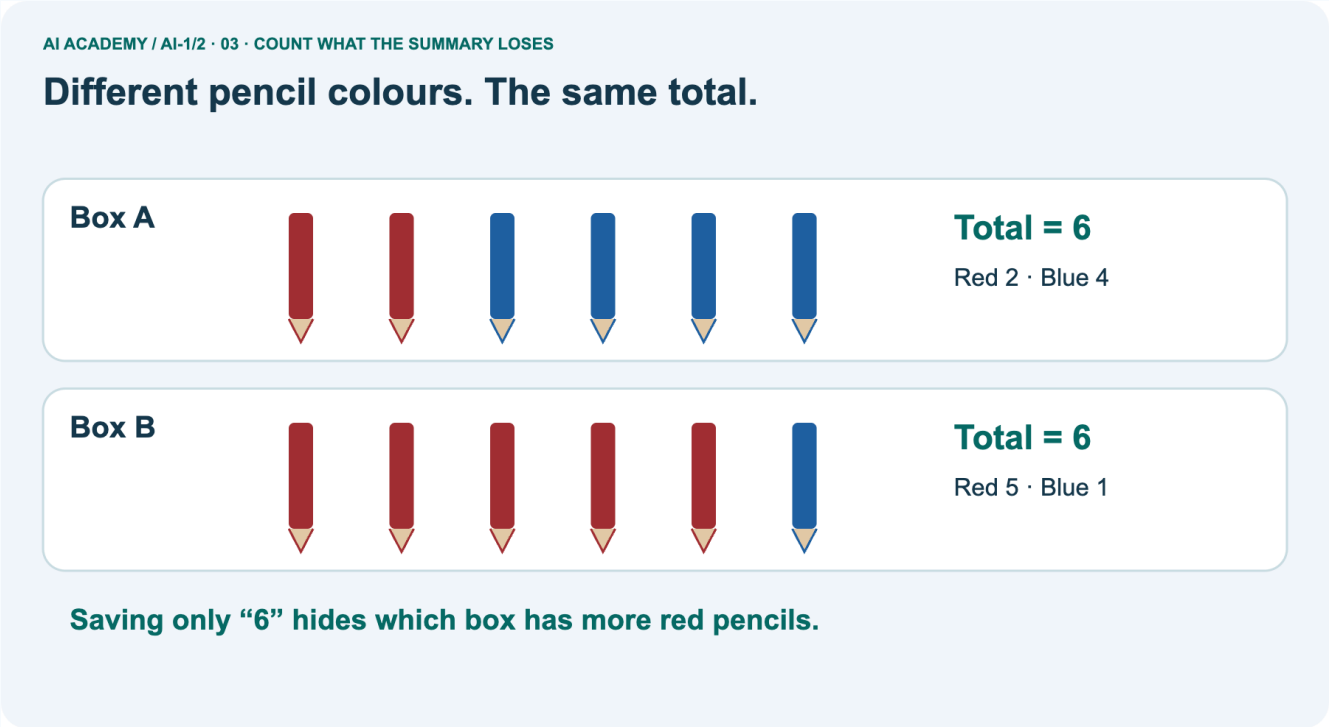


Follow the labelled picture with the matching explanation. Open the vector diagram

Read colours: A: red 2, blue 4. B: red 5, blue 1. → Add counts: A: 2+4=6. B: 5+1=6. → Store total: Both records become total 6. → Ask a new question: The saved totals cannot tell which box has more red pencils.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Different pencil colours. The same total.

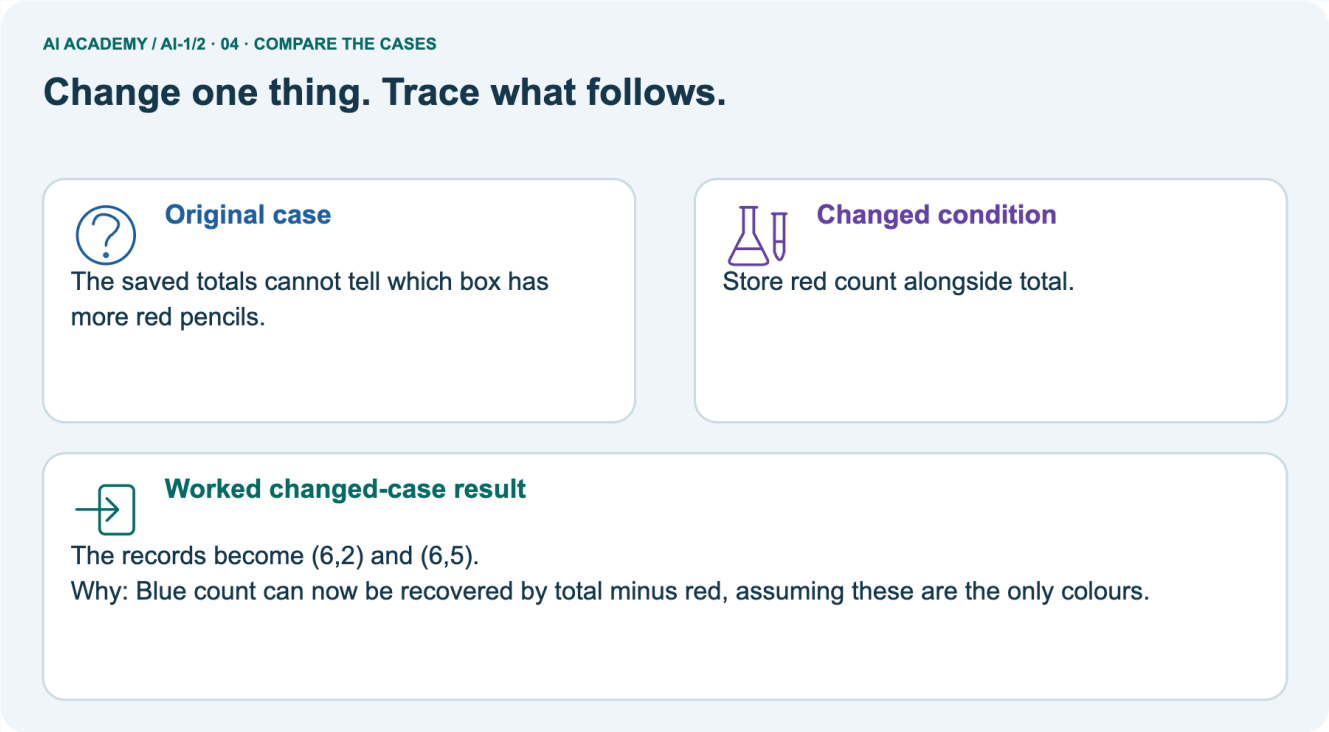


Follow the labelled picture with the matching explanation. Open the vector diagram

Box A: 2 red pencils and 4 blue, total 6. Box B: 5 red pencils and 1 blue, total 6. The equal totals do not preserve the colour counts.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



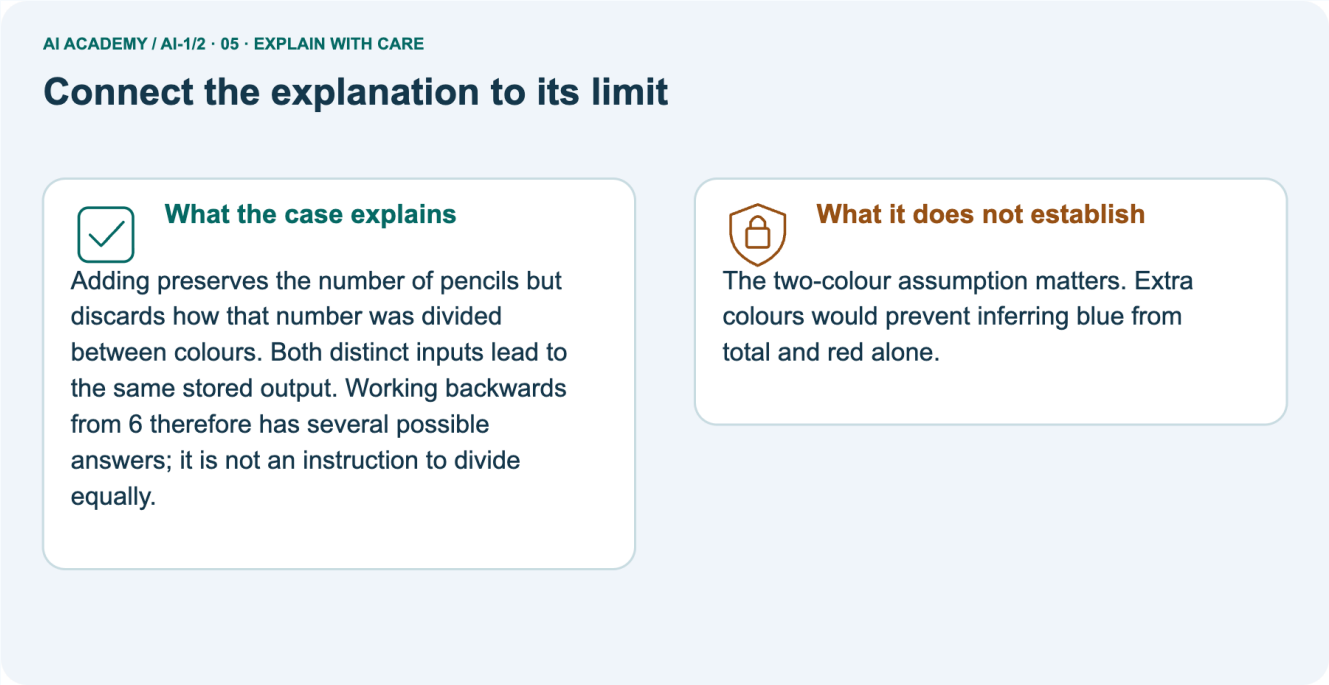
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The saved totals cannot tell which box has more red pencils. Changed condition: Store red count alongside total. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



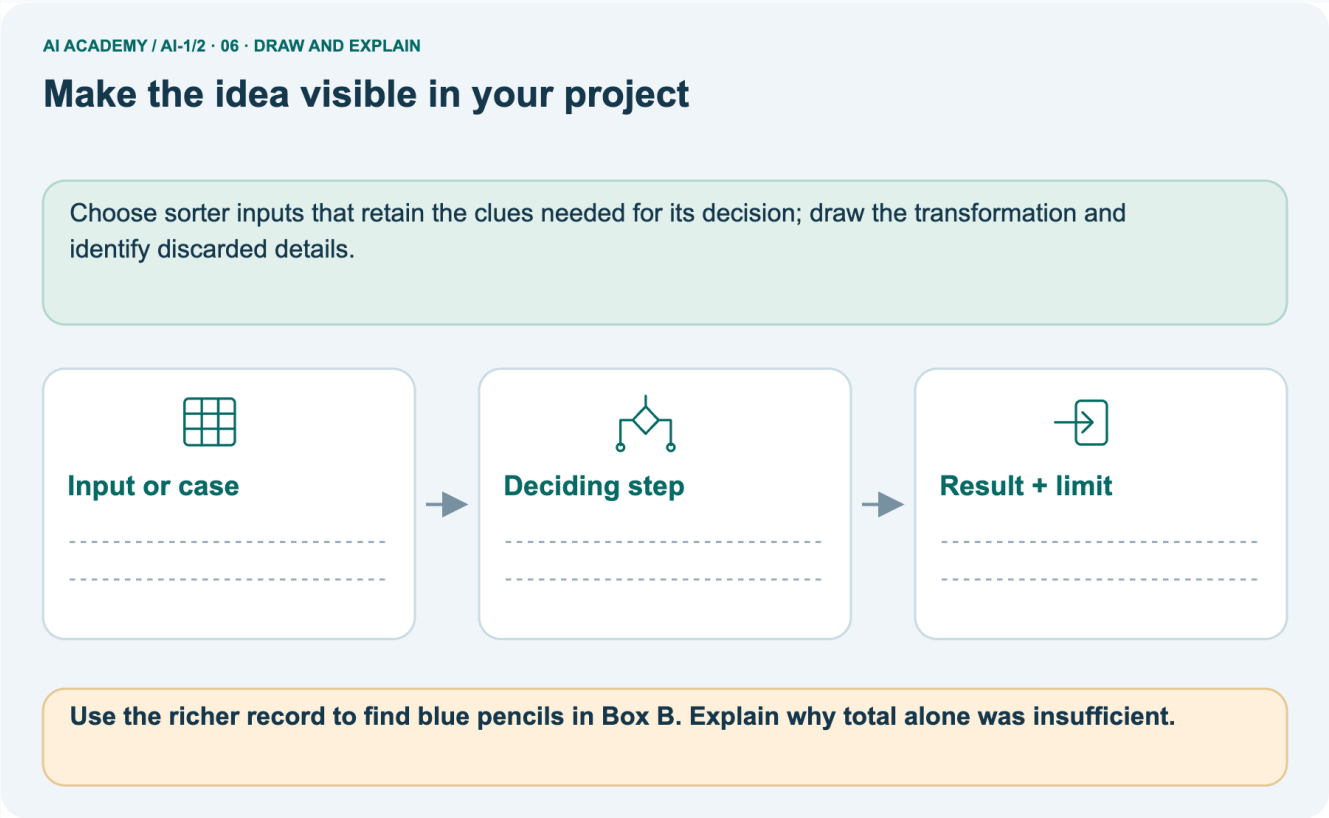
Follow the labelled picture with the matching explanation. Open the vector diagram

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally. Limit: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details. Draw your **input**, deciding step, result and limit.

Supported question: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

> Draw your project **trace** and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.